STUDENT INTERNSHIP-STYLE RESEARCH PROJECT

How can a consulting company grow without running short of cash?

A practical dashboard project using a fictional company, simple financial formulas and tested what-if experiments.

THE EXPERIMENT	I tested what happens when the company hires 40 people, customers pay later, utilization changes, or the future becomes more uncertain.

NovaBridge Digital SRL | Fictional Romanian consulting company | EUR

Prepared in August 2026 | 15-page student edition

All people, companies, invoices and results in this report are synthetic.

1 | WHY I CHOSE THIS TOPIC

A simple business problem with a cash timing twist

A consulting company pays salaries every month, but customers often pay invoices several weeks after the work is completed. Hiring can increase future revenue, yet salaries start before every employee is fully productive. Growth is therefore a cash question as well as a profit question.

RESEARCH QUESTION	Can the fictional company hire 40 people, earn more profit and still keep a comfortable cash balance while the new team ramps up?

My three goals

1. Understand revenue, operating profit and cash.
2. Build a dashboard where a beginner can change assumptions and see a result.
3. Test the model so a control cannot look useful while doing nothing.

The practical idea

I treated the dashboard as a small laboratory. Each page asks a question, lets the user change a few inputs, and explains what moved and why. The aim is learning, not pretending that fictional numbers are a real forecast.

2 | WHAT I BUILT

One project from data to conclusion

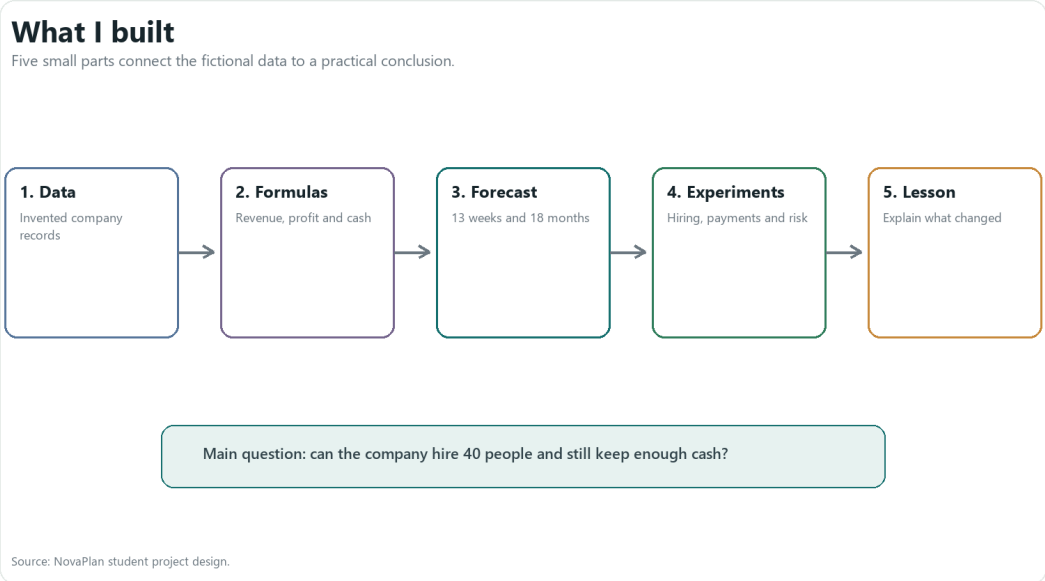


Figure 1. The project follows five steps instead of presenting many unrelated charts.

Dashboard page	Question
Project Overview	What problem am I studying?
What-if Experiment	What changes if I hire, sell more time, raise price or wait longer for cash?
13-Week Cash Check	Will the business have enough cash soon?
Risk Simulation	What range appears when the future is uncertain?
How It Works	Where do the data, formulas and tests come from?

3 | THE FICTIONAL DATA

Useful for learning, safe for a student project

Synthetic data is invented data made to follow selected rules. It lets me show invoices, salaries and cash movements without using confidential company or client information. Seed 42 makes the fictional history repeatable.



Figure 2. Four understandable data groups support the main story.

A USEFUL RULE The data is invented, but it must still add up: invoices match revenue, unpaid invoices become receivables, and opening cash plus cash movement equals closing cash.

4 | THE COMPANY STORY

Six years of growth, pressure and recovery

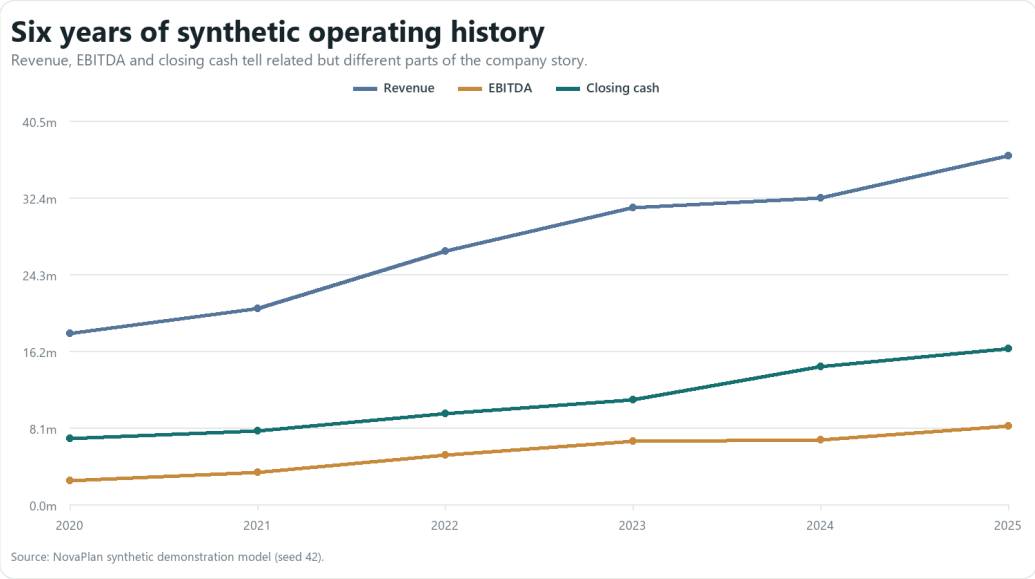


Figure 3. Revenue, operating profit and closing cash tell different parts of the 2020-2025 story.

2025 result	Value	Plain meaning
Revenue	EUR 36.8m	Services delivered
Operating profit	EUR 8.3m	Revenue left after main operating costs
Profit margin	22.5%	Operating profit divided by revenue
Closing cash	EUR 16.4m	Money available at year end

The company starts the forecast with a strong cash balance. This project is about funding growth, not rescuing a failing business.

5 | THREE IMPORTANT NUMBERS

Revenue, operating profit and cash are not the same

Number	Beginner definition	Small example
Revenue	Value of work completed	EUR 100 of work -> EUR 100 revenue
Operating profit	Revenue minus main operating costs	EUR 100 - EUR 75 -> EUR 25 profit
Cash	Money received and still available	An unpaid invoice is not cash yet

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Operating profit = Revenue - Payroll - Contractor cost - Other operating cost
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Closing cash = Opening cash + Cash collected - Cash paid
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THE KEY LESSON A profitable month can still use cash when customers have not paid yet, salaries are due now, or the company buys equipment.

Why the dashboard shows all three

Revenue explains business activity. Operating profit explains whether the work covers operating costs. Cash explains whether bills can be paid on time. Looking at only one gives an incomplete answer.

6 | HOW THE COMPANY MAKES MONEY

People and productive time create service revenue

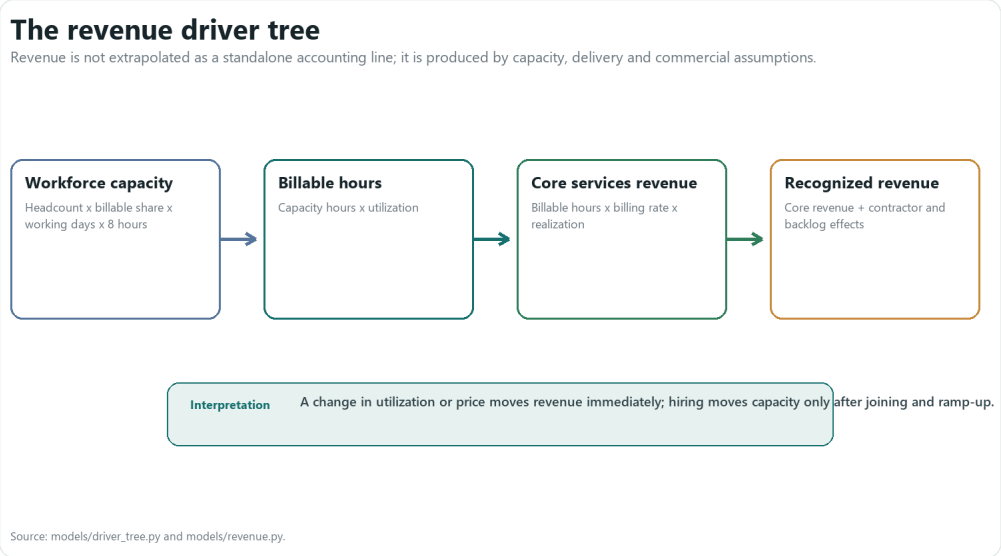


Figure 4. The revenue formula is a chain of understandable inputs.

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Revenue = Employees x Billable share x Available hours x Utilization x Hourly rate
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A worked example

Suppose 100 employees include 80 billable consultants. At 160 available hours, 75% utilization and EUR 75 per hour, the simple monthly estimate is 80 x 160 x 75% x EUR 75 = EUR 720,000 before discounts or other adjustments.

WHY UTILIZATION MATTERS	Hiring creates capacity, but capacity produces revenue only when customers buy the available time. New people do not create full revenue immediately.

7 | WHY PROFIT IS NOT CASH

Customer payment time creates the gap

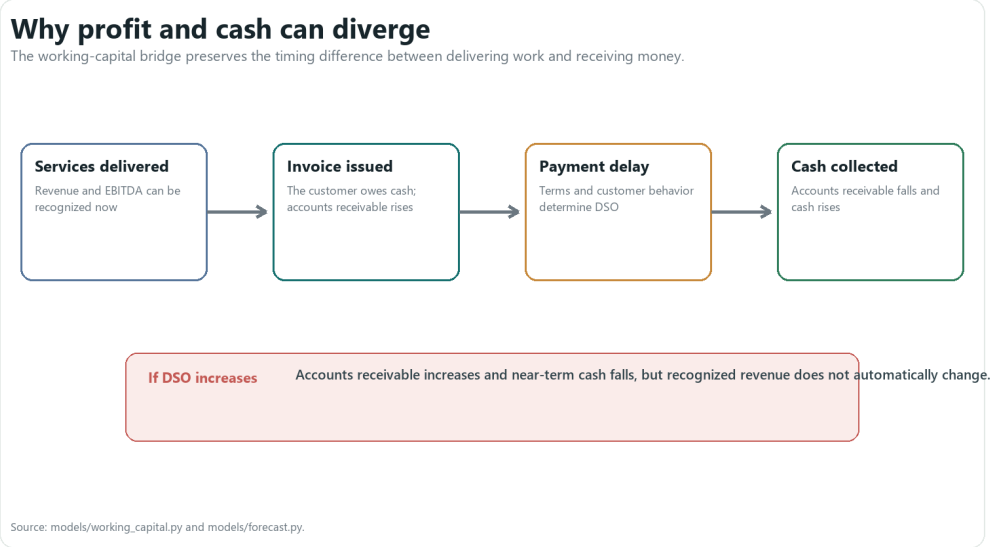


Figure 5. Revenue is recorded before cash when the customer pays later.

Term	Plain meaning
Invoice	A request for payment after work is delivered
Customer payment time	How long the company waits to collect
Accounts receivable	Invoices customers still owe
Cash collection	The moment money reaches the company

EXPERIMENT RULE Making customers pay 15 days later should reduce near-term cash, but not revenue or operating profit already earned.

8 | THE STARTING FORECAST

Two time horizons answer two questions

Forecast	Question
13 weeks	Can the company pay near-term salaries, suppliers, taxes and one-time bills?
18 months	How do hiring, utilization, prices and payments affect profit and cash?

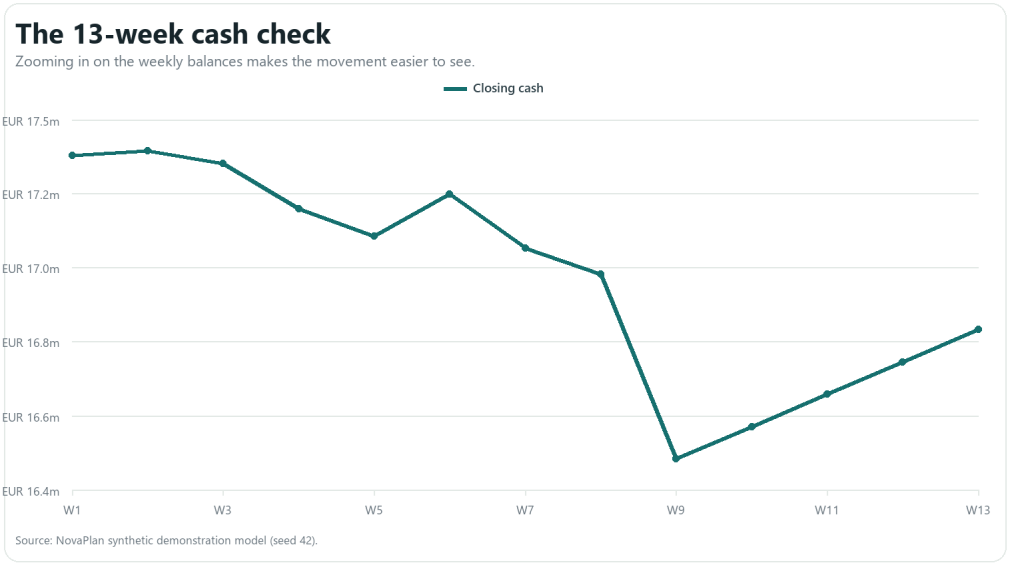


Figure 6. The weekly chart is zoomed so the cash movement is visible.

Cash check	Result
Opening cash	EUR 16.4m
Lowest weekly cash	EUR 16.5m
Cash after 13 weeks	EUR 16.8m

9 | EXPERIMENT 1

What happens if the company hires 40 people?

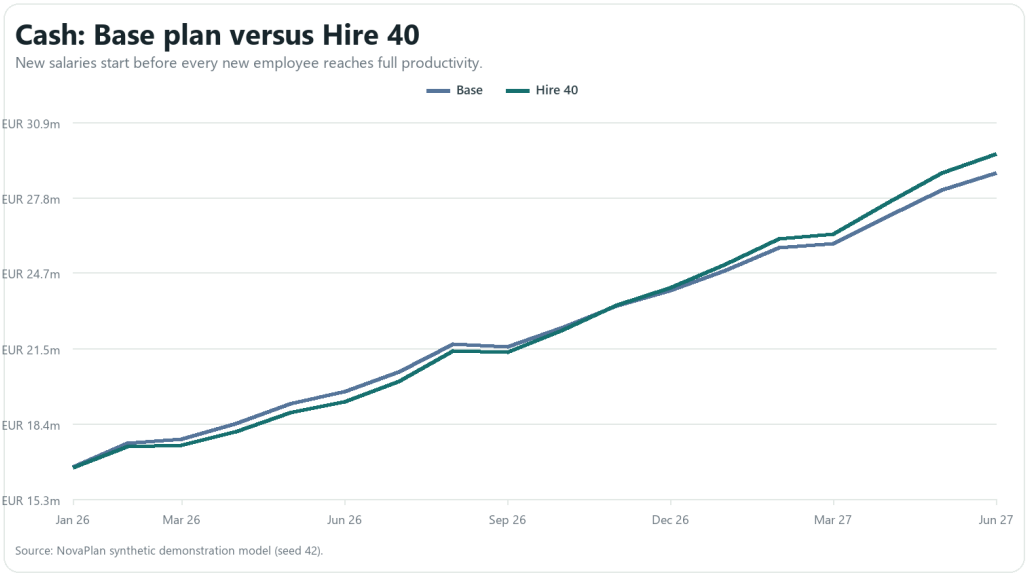


Figure 7. The report and dashboard use the same pure 40-hire experiment.

What changed	Result versus Base
Revenue	EUR 4.8m
Operating profit	EUR 1.7m
Lowest cash point	-EUR 43.6k
Employees at the end	492

WHAT I LEARNED

Hiring improves revenue and profit over 18 months, while the lowest cash point is slightly lower during ramp-up. This is a timing trade-off, not a simple good-or-bad answer.

10 | EXPERIMENT 2

Which dashboard input changes cash the most?

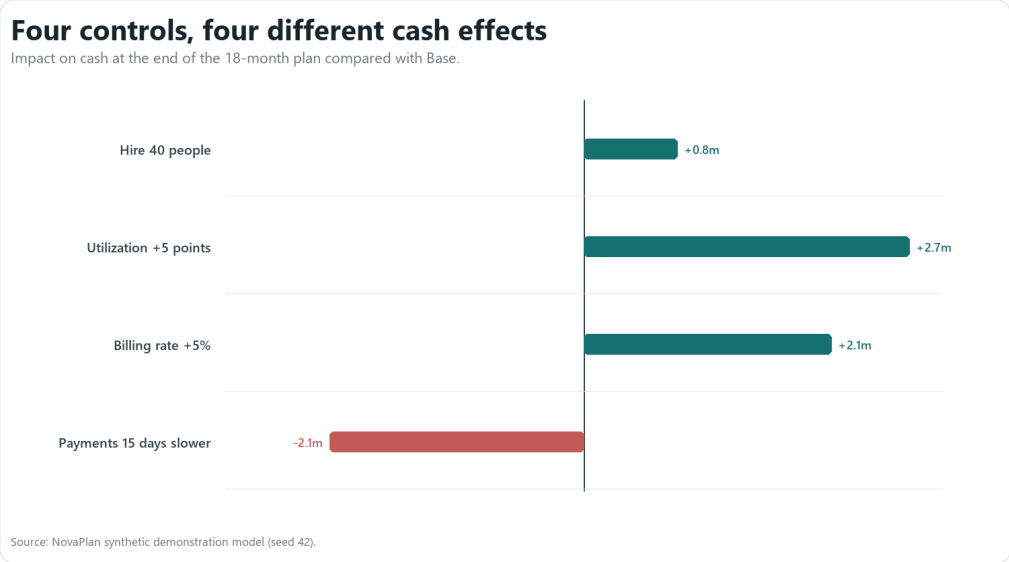


Figure 8. The four student-facing controls produce different end-of-horizon cash effects.

Experiment	Cash impact	Profit impact
Hire 40 people	EUR 787.6k	EUR 1.7m
Utilization +5 points	EUR 2.7m	EUR 3.8m
Billing rate +5%	EUR 2.1m	EUR 2.9m
Payments 15 days slower	-EUR 2.1m	EUR 0

PAYMENT-TIME CHECK Slower payment changes cash but leaves operating profit unchanged. This proves the model separates earning revenue from collecting it.

11 | A SMALL FORECASTING COMPARISON

Two simple ways to estimate next month

The financial plan uses business assumptions. I also compared two small forecasting ideas. Months stay in order so the model never learns from a future month before predicting it.

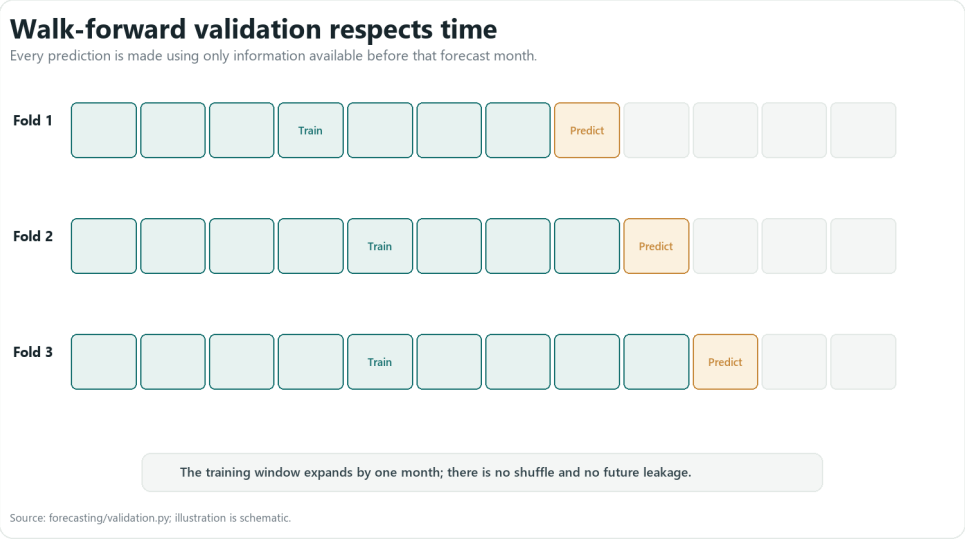


Figure 9. Walk-forward testing uses past months to predict the next month.

Method	How it works	Average revenue error
Seasonal repeat	Use the same month one year earlier	About EUR 330k
Simple driver regression	Use a straight-line relationship with planned drivers	About EUR 183k

HONEST LIMIT The regression assumes its driver values are already known or planned. It is a useful comparison, not an automatic forecast of the whole company.

12 | MONTE CARLO

A range of possible futures instead of one answer

A normal forecast gives one answer. Monte Carlo repeats the forecast while utilization, customer payment time, prices and salary costs move by small random amounts. Revenue, profit and cash are recalculated in every run.

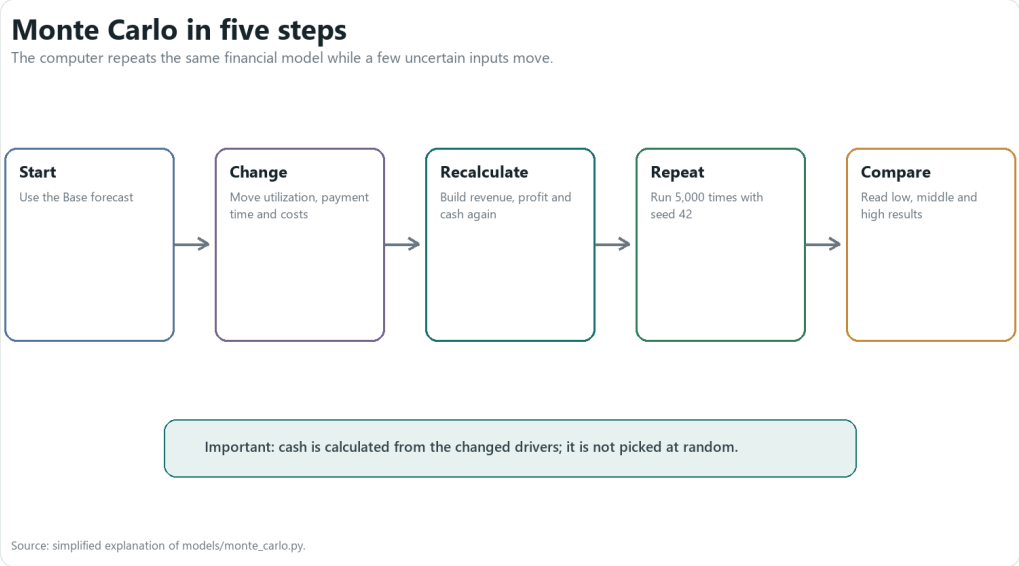


Figure 10. The calculation stays understandable even when repeated 5,000 times.

Result	Plain meaning
Low result (P5)	Only about 5% of ending cash results are lower
Middle result (P50)	Half are lower and half higher
High result (P95)	Only about 5% are higher

13 | TESTING MONTE CARLO

The controls now produce visible and logical changes

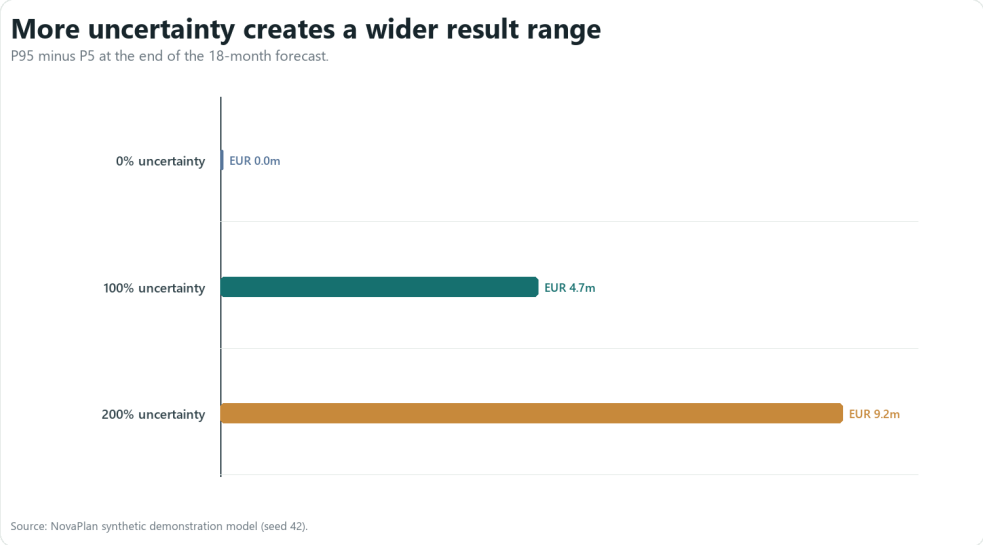


Figure 11. Zero uncertainty collapses the range; high uncertainty makes it much wider.

Uncertainty	P5	Middle	P95	Range
0%	EUR 28.8m	EUR 28.8m	EUR 28.8m	EUR 0
100%	EUR 26.5m	EUR 28.8m	EUR 31.2m	EUR 4.7m
200%	EUR 24.3m	EUR 28.8m	EUR 33.5m	EUR 9.2m

THE BUG I FOUND AND FIXED

The old page exposed controls that did not change business risk. The revised uncertainty control widens the paths. At 200% uncertainty, a EUR 14m target gives 0.8% below target and EUR 18m gives 98.4%, while the cash paths stay the same.

14 | WHAT I LEARNED

A useful model is one that can be explained and tested

ANSWER TO THE QUESTION	In this fictional Base case, the company can test a 40-person hiring plan without approaching the EUR 3m policy floor. The plan adds about EUR 1.7m of operating profit over 18 months, but salaries and customer payment timing still matter.

What I tested

- Revenue responds to utilization and billing rate.
- Slower payment reduces cash but not earned operating profit.
- Hiring includes salary cost and a gradual productivity ramp.
- The weekly cash roll-forward preserves receipts at forecast boundaries.
- Zero Monte Carlo uncertainty gives one path; more uncertainty widens the range.
- All 48 automated tests pass, every route loads, and the five main pages show no browser errors.

Limitations

The data is fictional, the business rules are simplified, and the simulated ranges are assumptions rather than proven probabilities. A future version could compare forecasts with later actual results and let users change a few uncertainty drivers separately.

Final reflection

The most important lesson was not a complicated algorithm. It was connecting a business action to revenue, profit and cash, then proving that each control really reaches the calculation.

EDUCATIONAL USE ONLY	NovaPlan is an educational proof-of-concept built using entirely synthetic data. It is not an official Deloitte product, does not represent Deloitte proprietary methods, and is not financial, tax, accounting or investment advice.